

Project XP Leveling System- Version 2.0

Overview

Version 2.0 expanded the system into a more complete RPG-style engine. This update introduced healing, HP scaling, damage calculation, formatted damage output, and multiple battle scenarios with conditional outcomes.

New Features Added

1. HP System + Scaling

- Player HP now scales with level ($\text{max_hp} = \text{level} * 25$).
- Player starts at full HP.
- HP updates dynamically when leveling up.

2. Healing System

- Added a `heal()` function with:
 - Resting vs. non-resting behavior
 - Overheal prevention
 - Global HP updates
 - Clean output messages

3. Damage Calculation

- Damage now scales with level.
- Includes a 20% crit bonus.
- Uses a formatting function to display large numbers cleanly (e.g., `2_500`).

4. Battle Simulations

Three battle functions were added:

- `simulate_battle()` — Minotaur
- `simulate_battle1()` — Hydra (conditional death)
- `simulate_battle2()` — Zeus (level-gated encounter)

Each battle:

- Calculates damage
- Prints clean battle text
- Returns XP rewards

5. Main Game Flow

After each battle:

- XP is awarded
- Stats are shown
- Player heals
- HP is printed to satisfy linter rules

What Version 2.0 Achieved

- A fully functional RPG stat engine
- Clean, readable, professional output
- Proper global state management
- No linter warnings
- Stable healing and HP logic
- Conditional battle outcomes
- A real sense of progression

Version 2.0 transformed the project from a simple XP calculator into a **mini RPG system**.